A factorized Pauli–Villars multigrid method for the Möbius domain-wall operator

Wei-Lun Chen

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The Möbius domain-wall operator has a spectrum that surrounds the origin, which defeats non-Hermitian Krylov solvers and most multigrid constructions. Kanamori, Chen and Matsufuru showed that preconditioning with the Pauli–Villars operator together with the site-diagonal part of the operator, $A' = (D_{\mathrm{PV}}C_{\mathrm{PV}}^{-1})^{\dagger}DC^{-1}$, moves the spectrum into the right half-plane and gives the smallest condition number among the operators they examined. This note describes a two-grid method for A'. The coarse-grid correction is factorized into one Galerkin solve for the quark operator and one for the adjoint Pauli–Villars operator, both with a nearest-neighbor coarse stencil, so that the outer system can be taken on the even sites without enlarging the coarse stencil; the prolongation is built from lifted approximate low modes for which the correction is exact on the vectors it is made from; the smoother is a fixed Chebyshev polynomial. We state the operator and its forms (with Neff's α), the correction, the coarse solvers, the setup, the smoother, the outer solver, and the questions that are open. On a quenched $24^3 \times 48$ configuration the multigrid iterations grow by a factor of 2–4 from $m = 0.1$ to $m = 0.001$, against 20 for even-odd CGNE, and at $m = 0.001$ the solve is 4.7–6.0 times faster than CGNE in double precision and 3.6–4.2 times faster than in mixed precision, the setup being paid back after 3–6 right-hand sides. Although the field of values of the preconditioned operator reaches far into the left half-plane, its eigenvalues stay at $\mathrm{Re}\,\lambda \geq 0.78$, and its pseudospectra on 12^4 and 16^4 bound the convergence of full GMRES; the truncated GCR used in practice converges only because the coarse solves are inexact. The results are from one configuration per volume; Sec. XII describes how to extend them.

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I. Introduction

Lattice computations of flavor physics and of the hadronic contributions to precision observables need a fermion discretization whose chiral symmetry is broken only mildly at finite lattice spacing. The overlap operator [1, 2] has an exact lattice chiral symmetry. Domain-wall fermions [3, 4, 5] approximate it at a lower cost by adding a fifth dimension of extent L_s, in which modes of opposite chirality are bound to the two boundaries by a large negative Wilson mass $-M_5$; the residual chiral symmetry breaking is suppressed as L_s grows, and the effective four-dimensional operator is a rational approximation of the overlap operator. The Möbius generalization [6, 7, 8] introduces two coefficients (b, c) and improves this approximation at fixed L_s; $c = 0$ is Shamir's operator.

The price is paid in the linear solver. The operator is neither Hermitian nor positive, one application costs L_s four-dimensional Wilson hops, and the standard solver is still the conjugate gradient method on the normal equation of the even-odd preconditioned operator, whose condition number is the square of that of the operator itself. For Wilson-type fermions this problem has been solved by multigrid: the low modes that slow down Krylov solvers are locally coherent [9], so a few of them, cut into blocks, span the low-mode space, and adaptive aggregation multigrid makes the cost nearly independent of the quark mass [10, 11, 12].

Multigrid for domain-wall fermions. The domain-wall operator has a spectrum that encloses the origin, because of the large negative Wilson mass. This violates the half-plane condition of non-Hermitian Krylov methods, whose convergence is then of the order of the system size [13, 14]. Boyle [14] lists four multigrid approaches that have had a measure of success, and they differ in the operator that is coarsened. Cohen *et al.* [15] coarsened the squared unpreconditioned operator, with blocks covering the whole fifth dimension, and noted that its coarse operator reaches two hops. HDCG [16] accelerates the best standard solver, CG on the normal equation of the red-black (even-odd) operator, and coarsens that squared Schur complement; its coarse stencil is next-to-next-to-next-to-nearest neighbor, 81 points for blocks of at least 4^4. HDCR [17] works instead directly on the nearest-neighbor Hermitian indefinite operator $\gamma_5 R_5 D$ at $c = 0$, so that the coarse operator stays nearest neighbor, explicitly to make re-evaluation along an HMC trajectory affordable. Brower, Clark, Howarth and Weinberg [19] pointed out that the Pauli–Villars preconditioned operator $D_{\mathrm{PV}}^{-1} D$, not D, carries the physical four-dimensional degrees of freedom, solved the system $D_{\mathrm{PV}}^{\dagger} D x = D_{\mathrm{PV}}^{\dagger} b$ with coarse operators built from the projected Wilson kernel, and demonstrated it in the two-dimensional Schwinger model. Boyle and Yamaguchi [18] compared these approaches for Shamir fermions; HDCR could be set up and solved faster than a single red-black CG solve on large volumes. Two of their observations bear on the present method: among red-black solvers, CG on the Schur complement $M_{ee} - M_{eo} M_{oo}^{-1} M_{oe}$ needed far fewer iterations than on the left-preconditioned $1 - M_{ee}^{-1} M_{eo} M_{oo}^{-1} M_{oe}$; and combining a red-black smoother with the HDCR coarse correction was not effective, which they attributed to the eigenvectors of the red-black operator and of $\gamma_5 R_5 D$ not being aligned.

Recent work by Boyle and collaborators. The multiple right-hand-side HDCG of Ref. [14] works on the red-black Hermitian operator $(M_{ee} - M_{eo} M_{oo}^{-1} M_{oe})^{\dagger} (M_{ee} - M_{eo} M_{oo}^{-1} M_{oe})$ with a flexible ADEF2 two-level conjugate gradient, a coarse space spanned by blocked low modes obtained with Chebyshev filters, and a stationary coarse solver (deflation plus a Chebyshev polynomial). Treating several right-hand sides at once with block CG and batched matrix products makes the coarse space sub-dominant, and gains of up to a factor of twenty over red-black CG are reported at physical quark mass. In a Lattice 2026 talk, Boyle and Oare [20] named two limitations of this approach—the cost of forming the 81-point coarse operator, and that a single solve on a new gauge field, as in HMC, does not profit from many right-hand sides—and presented a multigrid that directly coarsens the two-hop operator $D_{\mathrm{PV}}^{\dagger} D$ with blocks that contain the whole fifth dimension, which gives a 33-point coarse stencil. They emphasized that $D_{\mathrm{PV}}^{\dagger} D$ is non-normal: its field of values can extend into the left half-plane although its spectrum does not, the bound of Eisenstat, Elman and Schultz [21] for GCR is then lost, and they introduced a shift in the smoothers. The coarse solve dominated the cost; a three-level multiple right-hand-side version performs similarly to multiple right-hand-side HDCG, and an exact dense inverse on a further coarse level is planned.

Site-diagonal and Pauli–Villars preconditioning. Kanamori, Chen and Matsufuru [22] examined the spectra of several preconditioned Möbius operators. Combining the Pauli–Villars factor with the site-diagonal part C of the operator, they found that $A' = (D_{\mathrm{PV}} C_{\mathrm{PV}}^{-1})^{\dagger} D C^{-1}$ has eigenvalues with positive real part and the smallest condition number among the candidates, while replacing D_{PV} by a block-diagonal truncation brings back eigenvalues with negative real part. Independently, Neff's modification of the domain-wall matrix [23, 24] rescales the fifth dimension by a site-local factor Q and changes the conditioning of the even-odd system.

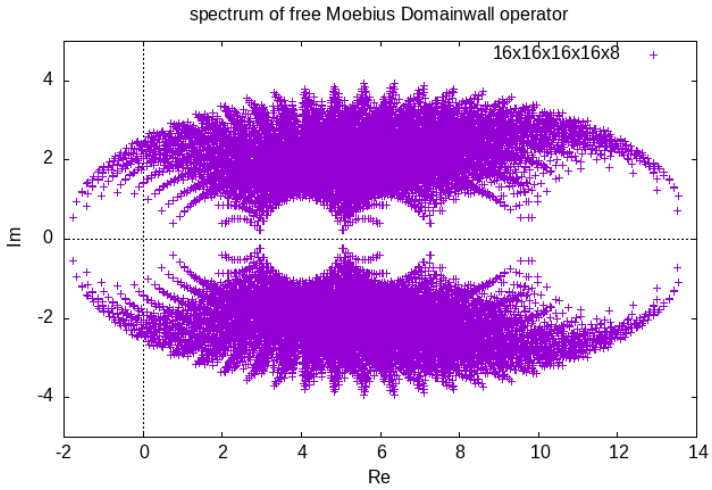


FIG. 1. Spectrum of a free Möbius domain-wall operator on $16^4 \times 8$ with quark mass 0.01, $b = 1.5$, $c = 0.5$ and domain-wall height $M_5 = 1$. Reproduced from Ref. [22], Fig. 1.

This note. We describe a two-grid method for A'. The outer system is the usual even-odd reduced one, as in HDCG, but the coarse-grid correction is *factorized*: the inverses of the quark operator and of the adjoint Pauli–Villars operator are replaced by two separate Galerkin inverses, each with a nearest-neighbor coarse stencil and an aggregation that is compatible with the five-dimensional chirality. The prolongation is built from approximate low modes lifted to the whole lattice in two ways, which makes the correction exact on these vectors for the right form of the operator. The smoother is a fixed Chebyshev polynomial and the setup combines Chebyshev filtering with a bootstrap. The note states the method and gives first numerical results: a mass scan against even-odd CGNE, the field of values, spectrum and pseudospectra of the preconditioned operator, the behavior of truncated GCR, and the sensitivity to the parameters.

Section II recalls the spectra on which the construction rests. Sections III–IX describe the method: the operator and its Pauli–Villars preconditioning, its even-odd forms, the factorized correction and the prolongation, the coarse solvers, the setup, the smoother and the outer solver. Section X sketches the implementation, Sec. XI gives the results, and Sec. XII how to reproduce and extend them. Section XIII lists the open problems. Statements that are our own derivations and have not been checked independently are marked **[derived here; not verified]**.

Conventions. Five-dimensional fields are $\psi_s(x)$, $s = 0, \ldots, L_s - 1$; $P_+, P_- = \frac{1}{2}(1 \pm \gamma_5)$. R reverses the fifth dimension, $(R\psi)_s = \psi_{L_s-1-s}$, and $\Gamma_5 = \gamma_5 R$. Subscripts e and o denote blocks on the even and odd four-dimensional sites; X_{pq} maps parity q to parity p. The injection ι puts a field on the even sites onto the whole lattice (zero on the odd sites), and $\pi = \iota^\dagger$ takes the even sites.

II. Spectra of preconditioned Möbius operators

The location of the spectrum decides which iterative methods can work. A Krylov residual polynomial p with $p(0) = 1$ that is small on the spectrum exists at low degree only if the spectrum can be enclosed by a region, such as an ellipse, that excludes the origin. The same holds for the Chebyshev smoother of Sec. VIII, which is defined by an ellipse in the right half-plane. The spectra of Ref. [22], reproduced in Figs. 1–3, show what the preconditioners do. The lattices are given in the captions.

The unpreconditioned operator. The free spectrum (Fig. 1) is a band that reaches into $\mathrm{Re}\,\lambda < 0$. On a $2 + 1$-flavor configuration at the physical point (Fig. 2, top left) the real parts lie in $[-1.29, 13.2]$. Conjugate gradients and BiCGStab therefore do not apply to D itself, and the practical choice has been

CG on a positive definite squared operator.

Site-diagonal preconditioning. With the site-diagonal part C, whose inverse is a constant matrix in the fifth dimension and spin (Sec. B), DC^{-1} has real parts in $[-0.22, 2.22]$ (middle left): the extent shrinks, the sign problem remains.

Pauli–Villars preconditioning. The operator $D_{\mathrm{PV}}^{-1}D$, with $D_{\mathrm{PV}} = D(m = 1)$, is a rational approximation of the overlap operator [19]. Because D_{PV}^{-1} is as expensive as a solve with D, Ref. [19] used $D_{\mathrm{PV}}^{\dagger}$ instead. $D_{\mathrm{PV}}^{\dagger}D$ has positive real parts (top right), and its low modes lie close to the imaginary axis with a positive shift of the order of the quark mass, as expected of an approximate overlap operator; its high end, however, reaches about 187, which an overlap operator would not show. Replacing D_{PV} by its block-diagonal approximation B_{PV}, which drops the hops between 4^4 blocks, brings back negative real parts (middle and bottom right).

Both together. The operator $A' = (D_{\mathrm{PV}}C_{\mathrm{PV}}^{-1})^{\dagger}DC^{-1}$ has real parts in $[2.67 \times 10^{-4},\ 5.22]$ (Fig. 3, left); its condition number, 1.96×10^4, is smaller than the 2.34×10^4 of $D_{\mathrm{PV}}^{\dagger}D$, and the smallest among the operators examined in Ref. [22]. With D_{PV} replaced by B_{PV} the real parts become negative again (right). A preconditioner that truncates the Pauli–Villars factor to blocks would therefore inherit eigenvalues with negative real part; in the method below the factor is kept exact on the fine lattice and is approximated only through its Galerkin coarse operator (Sec. V).

Two caveats. First, the figures show the full-lattice operator A'; the even-odd operators of Sec. IV are related to it but their spectra are not part of these figures. Second, positivity of the real parts of the eigenvalues is not positivity of the field of values. For a non-normal operator the convergence bound of GCR [21] requires the Hermitian part $(A + A^{\dagger})/2$ to be positive definite, which can fail while every eigenvalue lies in the right half-plane; this is the mechanism discussed for $D_{\mathrm{PV}}^{\dagger}D$ in Ref. [20]. Whether it occurs for A' and its even-odd forms is open (Sec. XIII). For $\tilde{A}_R$ itself the field of values reaches just below zero on 12^4 ($\mathrm{Re}\, W \geq -4.6 \times 10^{-3}$ with stout smearing, -5.0×10^{-2} without), and for the multigrid-preconditioned $\tilde{A}_R\mathcal{M}$ far below (Sec. C).

III. The operator and Pauli–Villars preconditioning

A. The Möbius operator

Let K be the hopping term of the Wilson operator,

$$(K\psi)(x) = \sum_{\mu=1}^{4} \Big[(1-\gamma_\mu)U_\mu(x)\psi(x+\hat{\mu}) + (1+\gamma_\mu)U_\mu^{\dagger}(x-\hat{\mu})\psi(x-\hat{\mu})\Big], \tag{3.1}$$

and $W = (4 - M_5) - \frac{1}{2}K$ the Wilson operator with mass $-M_5$ and hopping coefficient $\frac{1}{2}$. The hop in the fifth dimension is

$$(Y\psi)_s = -P_-\psi_{s+1} - P_+\psi_{s-1}, \qquad \psi_{L_s} \equiv -m\,\psi_0, \quad \psi_{-1} \equiv -m\,\psi_{L_s-1}, \tag{3.2}$$

where m is the quark mass, which enters only through the wrap-around. The Möbius operator is

$$D = 1 + Y + W\,(b - cY) = C - \tfrac{1}{2}K\,(b - cY), \qquad C = d_+ + d_-Y, \tag{3.3}$$

with $d_+ = 1 + b(4 - M_5)$ and $d_- = 1 - c(4 - M_5)$. K acts on each slice s and does not commute with P_+ and P_-, so the order in Eq. (3.3) matters. The Pauli–Villars operator is the same operator at unit mass, $D_{\mathrm{PV}} = D|_{m=1}$, $C_{\mathrm{PV}} = C|_{m=1}$.

Γ_5-*Hermiticity.* Since $\Gamma_5 K \Gamma_5 = K^\dagger$, $\Gamma_5 Y \Gamma_5 = Y^\dagger$ and $\Gamma_5 C \Gamma_5 = C^\dagger$,

$$\Gamma_5 D \Gamma_5 - D^\dagger = -\tfrac{1}{2}\big[K^\dagger,\, b - cY^\dagger\big] = \tfrac{c}{2}\big[K^\dagger, Y^\dagger\big]. \tag{3.4}$$

The commutator is non-zero because K contains γ_μ and Y contains $P_\pm$. Thus $\Gamma_5 D \Gamma_5 = D^\dagger$ holds for Shamir's operator and fails at $O(c)$ for $c \neq 0$; the Hermitian operator of the Möbius case is $\Gamma_5 D_-^{-1} D$ with $D_- = 1 - cW$ [8], which involves an inverse that is not local. This matters for the coarse solvers (Sec. VI).

B. The site-diagonal part and its inverse

C acts within a four-dimensional site, as an $L_s \times L_s$ matrix in the fifth dimension on each chirality. With $\psi^\pm = P_\pm \psi$, it is lower bidiagonal on ψ^+ and upper bidiagonal on ψ^-, each with one corner element from the mass:

$$(C\psi)^+_s = \begin{cases} d_+ \psi^+_0 + m\, d_- \psi^+_{L_s-1}, & s = 0, \\ d_+ \psi^+_s - d_- \psi^+_{s-1}, & s \geq 1, \end{cases} \tag{3.5}$$

and the same for ψ^- with $s \to L_s - 1 - s$. With $q = d_-/d_+$, the solution of $C^+ x = r$ is a forward substitution $p_0 = 0$, $p_s = (r_s + d_- p_{s-1})/d_+$, followed by

$$x_0 = \frac{r_0 - m\, d_-\, p_{L_s-1}}{d_+ + m\, d_-\, q^{L_s-1}}, \qquad x_s = p_s + q^s x_0; \tag{3.6}$$

for C^- the substitution runs from $s = L_s - 1$ downwards, and for $C^\dagger$ the roles of the two chiralities are exchanged. C^{-1} is thus exact and costs $O(L_s)$ per site. It does not depend on the gauge field [22].

C. The preconditioned operator

The operator of Ref. [22] is

$$A' = (D_{\rm PV} C_{\rm PV}^{-1})^\dagger\, D\, C^{-1}, \qquad A' y = (D_{\rm PV} C_{\rm PV}^{-1})^\dagger\, \eta, \quad \psi = C^{-1} y, \tag{3.7}$$

for $D\psi = \eta$. For $m \to 1$ it becomes the Hermitian positive operator $(D_{\rm PV} C_{\rm PV}^{-1})^\dagger (D_{\rm PV} C_{\rm PV}^{-1})$.

D. Neff's α

Neff's form of the domain-wall matrix [23, 24] is $D^{(\alpha)} = DQ$ with a site-local Q, diagonal in the fifth dimension,

$$Q_0 = P_- + \alpha P_+, \qquad Q_s = \alpha \;\; (0 < s < L_s - 1), \qquad Q_{L_s-1} = P_+ + \alpha P_-. \tag{3.8}$$

Q is Hermitian, Q^{-1} is Eq. (3.8) with $\alpha \to 1/\alpha$, and Q leaves $P_-\psi_0$ and $P_+\psi_{L_s-1}$ unchanged. The site-diagonal part of DQ is CQ, with $(CQ)^{-1} = Q^{-1}C^{-1}$ and $(CQ)^\dagger = QC^\dagger$. The solution of $D\psi = \eta$ is $\psi = Q\psi^{(\alpha)}$ with $D^{(\alpha)}\psi^{(\alpha)} = \eta$. The factor may be applied to the quark operator alone or to the Pauli–Villars operator as well; we write $D^{(\alpha)}_{\rm PV} = D_{\rm PV} Q$ in the latter case.

IV. Even-odd forms

With $D = \begin{pmatrix} C_e & D_{eo} \\ D_{oe} & C_o \end{pmatrix}$, $D_{pq} = -\tfrac{1}{2} K_{pq}(b - cY)$, the Schur complement on the even sites is $S = C_e - D_{eo} C_o^{-1} D_{oe}$, and $S_{\rm PV}$ is that of $D_{\rm PV}$; note $S^{-1} = \pi D^{-1} \iota$ and $(S^\dagger_{\rm PV})^{-1} = \pi (D^\dagger_{\rm PV})^{-1} \iota$. The site-diagonal factor can stand on either side of S, which gives the right and the left form,

$$\tilde{A}_R = (S_{\rm PV} C^{-1}_{{\rm PV},e})^\dagger\, (S\, C_e^{-1}), \qquad \tilde{A}_L = (C^{-1}_{{\rm PV},e} S_{\rm PV})^\dagger\, (C_e^{-1} S). \tag{4.1}$$

TABLE 1. Site-local factors of the correction (5.2). They follow from $\tilde{A}^{-1}$ with $S^{-1} = \pi D^{-1}\iota$; with α the site-diagonal blocks are C_eQ_e and $Q_eC_{\mathrm{PV},e}^\dagger$.

form	O_R	O_M	O_L
right, any α	$C_{\mathrm{PV},e}^\dagger$	1	C_e
left, α in D only	1	$C_eQ_eC_{\mathrm{PV},e}^\dagger$	Q_e^{-1}
left, α in D and D_{PV}	Q_e^{-1}	$C_eQ_e^2C_{\mathrm{PV},e}^\dagger$	Q_e^{-1}

$\tilde{A}_R$ is the even-odd counterpart of A'; $C_e^{-1}S$ is the usual left preconditioned even-odd operator, which Boyle and Yamaguchi found inferior to S itself for CG on the normal equation [18]. With α, DQ has Schur complement SQ_e and diagonal block C_eQ_e, so $\tilde{A}_R$ is independent of α, whereas

$$\tilde{A}_L^{(\alpha)} = \begin{cases} Q_e\,(C_{\mathrm{PV},e}^{-1}S_{\mathrm{PV}})^\dagger\,Q_e^{-2}\,(C_e^{-1}S)\,Q_e, & \alpha \text{ in } D \text{ and } D_{\mathrm{PV}}, \\ (C_{\mathrm{PV},e}^{-1}S_{\mathrm{PV}})^\dagger\,Q_e^{-1}\,(C_e^{-1}S)\,Q_e, & \alpha \text{ in } D \text{ only}. \end{cases} \tag{4.2}$$

For $D\psi = \eta$, with $\eta'_e = \eta_e - D_{eo}C_o^{-1}\eta_o$, the right-hand side is $\rho_R = (S_{\mathrm{PV}}C_{\mathrm{PV},e}^{-1})^\dagger\eta'_e$ or $\rho_L = (C_{\mathrm{PV},e}^{-1}S_{\mathrm{PV}})^\dagger C_e^{-1}\eta'_e$ (operators with Q when $\alpha \neq 1$); the solution is $\psi_e^{(\alpha)} = C_e^{-1}y$ (right) or y (left), $\psi_o^{(\alpha)} = C_o^{-1}(\eta_o - D_{oe}\psi_e^{(\alpha)})$, $\psi = Q\psi^{(\alpha)}$.

V. The factorized coarse-grid correction and the prolongation

A. Why factorize

Solving on the even sites halves the problem and improves the conditioning, but in the literature it has come at a price: the Schur complement is a two-hop operator, so HDCG, which coarsens the squared Schur complement, has an 81-point coarse stencil [16], while the methods with a more compact coarse operator coarsen something other than the Schur complement: γ_5R_5D (HDCR [17]) or $D_{\mathrm{PV}}^\dagger D$ [19, 20]. The factorized correction avoids this choice: the outer operator is the even-odd one, while the coarse operators are those of D and D_{PV} themselves. By $S^{-1} = \pi D^{-1}\iota$, the inverse of the right form is

$$\tilde{A}_R^{-1} = C_e\,S^{-1}\,(S_{\mathrm{PV}}^\dagger)^{-1}\,C_{\mathrm{PV},e}^\dagger = C_e\,\pi D^{-1}\iota\;\pi(D_{\mathrm{PV}}^\dagger)^{-1}\iota\;C_{\mathrm{PV},e}^\dagger, \tag{5.1}$$

a product of two inverses of one-hop operators with site-local factors. The Galerkin operator of $\tilde{A}_R$ itself, a four-hop operator on the fine lattice, would have a much larger coarse stencil; the Galerkin operators of D and D_{PV} are nearest-neighbor, because D is nearest-neighbor in four dimensions and every aggregate contains the whole fifth dimension. We therefore replace each inverse by its Galerkin approximation $\mathcal{P}(\mathcal{P}^\dagger X\mathcal{P})^{-1}\mathcal{P}^\dagger$. For both forms and any α, the correction of a residual r on the even sites is

$$e = O_L\;\pi\mathcal{P}\,\mathcal{D}_{\mathrm{c}}^{-1}\mathcal{P}^\dagger\iota\;\;O_M\;\;\pi\mathcal{P}\left(\mathcal{P}^\dagger D_{\mathrm{PV}}^\dagger\mathcal{P}\right)^{-1}\mathcal{P}^\dagger\iota\;\;O_R\,r, \qquad \mathcal{D}_{\mathrm{c}} = \mathcal{P}^\dagger D\mathcal{P}, \tag{5.2}$$

with the site-local factors of Table 1. The coarse operators are always those of the plain operators D and D_{PV} (no α), so one setup serves both forms and every α.

B. Aggregation and prolongation

The fine lattice is divided into rectangular four-dimensional blocks; each block contains the whole fifth dimension and becomes one coarse site. Each test vector v is split into its Γ_5 eigencomponents $v_\pm = \frac{1}{2}(1\pm\Gamma_5)v$, and within each block the $+$ parts and the $-$ parts are orthonormalized separately. With n_v test vectors a coarse site carries $2n_v$ components, the first n_v of positive and the last n_v of negative Γ_5-chirality. Writing $\Sigma = \mathrm{diag}(\mathbf{1}_{n_v}, -\mathbf{1}_{n_v})$ on each coarse site,

$$\Gamma_5\mathcal{P} = \mathcal{P}\Sigma, \tag{5.3}$$

so that $\mathcal{P}^\dagger(\Gamma_5 X \Gamma_5)\mathcal{P} = \Sigma\,\mathcal{P}^\dagger X \mathcal{P}\,\Sigma$ for any fine operator X; in particular $\mathcal{P}^\dagger D_{\mathrm{PV}}^\dagger \mathcal{P} = \Sigma \mathcal{D}_{\mathrm{c}}^{\mathrm{PV}} \Sigma$ with $\mathcal{D}_{\mathrm{c}}^{\mathrm{PV}} = \mathcal{P}^\dagger D_{\mathrm{PV}} \mathcal{P}$ whenever $\Gamma_5 D_{\mathrm{PV}} \Gamma_5 = D_{\mathrm{PV}}^\dagger$, i.e. for $c = 0$. Because $v_\pm(L_s - 1 - s) = \pm\gamma_5 v_\pm(s)$, each part is fixed by half of the fifth dimension, so $\mathcal{P}$ need only be stored for $s < L_s/2$. On the even sites the transfer operators are $\mathcal{P}^\dagger\iota$ and $\pi\mathcal{P}$.

C. The basis $\{s, \hat{t}\}$

The prolongation is built from vectors y_j on the even sites that approximate low modes of $\tilde{A}_R$ (Sec. VII). Each y is lifted in two ways:

$$s = \big(C_e^{-1} y,\ -C_o^{-1} D_{oe} C_e^{-1} y\big), \qquad Ds = \iota\, S C_e^{-1} y, \tag{5.4}$$

$$\hat{t} = \big(a,\ -(C_{\mathrm{PV},o}^\dagger)^{-1} (D_{\mathrm{PV},eo})^\dagger a\big), \quad a = \pi D s, \qquad D_{\mathrm{PV}}^\dagger \hat{t} = \iota\, S_{\mathrm{PV}}^\dagger a, \tag{5.5}$$

where the right-hand columns follow from the block form of D and of $D_{\mathrm{PV}}^\dagger$, whose odd-even block is $(D_{\mathrm{PV},eo})^\dagger$ and whose odd diagonal block is $C_{\mathrm{PV},o}^\dagger$. The images of s under D and of $\hat{t}$ under $D_{\mathrm{PV}}^\dagger$ thus vanish on the odd sites. The $2n$ vectors $\{s_j, \hat{t}_j\}$ are the test vectors, $n_v = 2n$.

Proposition 1 (Exactness, right form)**.** *Let $s, \hat{t} \in \operatorname{range}\mathcal{P}$, let $\mathcal{D}_{\mathrm{c}}$ and $\mathcal{P}^\dagger D_{\mathrm{PV}}^\dagger \mathcal{P}$ be invertible, and let both coarse systems in Eq. (5.2) be solved exactly with the exact adjoint $\mathcal{P}^\dagger D_{\mathrm{PV}}^\dagger \mathcal{P} = (\mathcal{D}_{\mathrm{c}}^{\mathrm{PV}})^\dagger$. Then the right-form correction of $r = \tilde{A}_R y$ is $e = y$.*

Proof. For any invertible $\mathcal{P}^\dagger X \mathcal{P}$, $\mathcal{P}(\mathcal{P}^\dagger X \mathcal{P})^{-1}\mathcal{P}^\dagger X$ is the identity on $\operatorname{range}\mathcal{P}$. Now $O_R r = C_{\mathrm{PV},e}^\dagger \tilde{A}_R y = S_{\mathrm{PV}}^\dagger S C_e^{-1} y = S_{\mathrm{PV}}^\dagger a$, and $\iota S_{\mathrm{PV}}^\dagger a = D_{\mathrm{PV}}^\dagger \hat{t}$ by Eq. (5.5). The first coarse solve returns $\pi\hat{t} = a = \pi D s$. With $O_M = 1$ the second has $\mathcal{P}^\dagger \iota \pi D s = \mathcal{P}^\dagger D s$ on the right, since Ds vanishes on the odd sites, and returns $\pi s = C_e^{-1} y$; $O_L = C_e$ gives y. □

Remark 1 (Conditions in practice). **[derived here; not verified]** The proposition needs (i) $s, \hat{t}$ in the range of the stored $\mathcal{P}$, which holds only up to the rounding of the reduced-precision storage of $\mathcal{P}$; (ii) the exact adjoint of the Pauli–Villars coarse factor—the Σ-adjoint $\Sigma\mathcal{D}_{\mathrm{c}}^{\mathrm{PV}}\Sigma$ of Sec. VI equals it only if $\Gamma_5 D_{\mathrm{PV}} \Gamma_5 = D_{\mathrm{PV}}^\dagger$, which by Eq. (3.4) fails for $c \neq 0$; (iii) exact coarse solves, whereas they are stopped at a loose tolerance. How much each departure costs has not been analyzed. Numerically (ii) costs nothing measurable (Sec. G), and (iii) helps: with GCR(2), accurate coarse solves need more iterations or stagnate (Secs. F, H).

Remark 2 (Left form). **[derived here; not verified]** For the left form with α in both operators, the same argument requires the quark lift s' of $Q_e y'$ and the Pauli–Villars lift $\hat{t}'$ of $u = C_{\mathrm{PV},e}^{\dagger\,-1} Q_e^{-2} C_e^{-1} S Q_e y'$ to lie in $\operatorname{range}\mathcal{P}$ (Table 1 gives $O_M u = S Q_e y'$). For $\alpha = 1$ and $y' = C_e^{-1} y$ one has $s' = s$, but $u = C_{\mathrm{PV},e}^{\dagger\,-1} C_e^{-1} a$ differs from a by a site-local matrix that mixes s and spin, so $\hat{t}'$ is in general not in the span of the right-form vectors. We therefore expect the correction not to be exact for the left form with the basis above, but this has not been verified. In practice the left form with this basis has the spectrum of the right form and needs one to three iterations more or fewer (Secs. D, F).

D. The coarse operators

$\mathcal{D}_{\mathrm{c}} = \mathcal{P}^\dagger D \mathcal{P}$ and $\mathcal{D}_{\mathrm{c}}^{\mathrm{PV}} = \mathcal{P}^\dagger D_{\mathrm{PV}} \mathcal{P}$ are assembled from the hops of the fine operator between neighboring blocks and from its block-internal part: on each coarse site a $2n_v \times 2n_v$ diagonal block and one link block per direction, nine blocks in all. The adjoint factor $\mathcal{P}^\dagger D_{\mathrm{PV}}^\dagger \mathcal{P}$ is not assembled; it is applied either exactly, as the conjugate transpose $(\mathcal{D}_{\mathrm{c}}^{\mathrm{PV}})^\dagger$ of the stored blocks, or approximately as $\Sigma\mathcal{D}_{\mathrm{c}}^{\mathrm{PV}}\Sigma$. When all extents of the coarse lattice are even, the coarse hops connect coarse sites of opposite parity and both coarse operators admit an even-odd decomposition of their own.

VI. The coarse solvers

Both coarse systems are solved approximately, to a loose relative residual or a maximum number of iterations, in reduced precision; reaching the tolerance is not required. They are solved in sequence: first the Pauli–Villars factor, whose solution enters the right-hand side of the quark factor.

A. The quark factor

Write $\mathcal{D}_{\mathrm{c}} = \begin{pmatrix} A_{ee} & B_{eo} \\ B_{oe} & A_{oo} \end{pmatrix}$ on the coarse lattice. The system $\mathcal{D}_{\mathrm{c}} c = g$ is solved through its Schur complement,

$$\left(A_{ee} - B_{eo} A_{oo}^{-1} B_{oe}\right) c_e = g_e - B_{eo} A_{oo}^{-1} g_o, \qquad c_o = A_{oo}^{-1}(g_o - B_{oe} c_e), \tag{6.1}$$

with BiCGStab or truncated flexible GCR, preconditioned by the exact inverses of the diagonal blocks A_{ee}^{-1}. $\mathcal{D}_{\mathrm{c}}$ is not Hermitian and $\Sigma\mathcal{D}_{\mathrm{c}}$ is Hermitian only for $c = 0$, so a non-Hermitian Krylov method is used.

B. The Pauli–Villars factor

The system is $(\mathcal{P}^\dagger D_{\mathrm{PV}}^\dagger \mathcal{P})\, w = g$. Two treatments are available.

Exact adjoint. Solve $(\mathcal{D}_{\mathrm{c}}^{\mathrm{PV}})^\dagger w = g$ with a non-Hermitian Krylov method, preconditioned by the inverses of the adjoint diagonal blocks, $(A_{xx}^\dagger)^{-1} = (A_{xx}^{-1})^\dagger$.

Σ-Hermitian form. Replace the adjoint by $\Sigma\mathcal{D}_{\mathrm{c}}^{\mathrm{PV}}\Sigma$ and solve the equivalent system

$$\mathcal{H}_{\mathrm{c}}\, w = \Sigma g, \qquad \mathcal{H}_{\mathrm{c}} = \mathcal{D}_{\mathrm{c}}^{\mathrm{PV}}\Sigma, \tag{6.2}$$

with MINRES. $\mathcal{H}_{\mathrm{c}}$ is Hermitian (indefinite) exactly when $(\mathcal{D}_{\mathrm{c}}^{\mathrm{PV}})^\dagger = \Sigma\mathcal{D}_{\mathrm{c}}^{\mathrm{PV}}\Sigma$, i.e. for $c = 0$ by Eqs. (3.4) and (5.3); for $c \neq 0$ MINRES is applied to a non-Hermitian matrix and has no guarantee. In practice it is better than the exact adjoint with BiCGStab for $c = 0.5$ as for $c = 0$ (Sec. G).The preconditioner must be Hermitian positive definite; we use the absolute value of the diagonal blocks [25], $|\mathcal{H}_{xx}|^{-1} = V|\Lambda|^{-1}V^\dagger$ with $\frac{1}{2}(\mathcal{H}_{xx} + \mathcal{H}_{xx}^\dagger) = V\Lambda V^\dagger$ on each coarse site. On a coarse lattice with even extents, the Schur complement of $\mathcal{H}_{\mathrm{c}}$ on the even sites is $S_{\mathrm{c}}\Sigma_e$, with S_{c} the Schur complement of $\mathcal{D}_{\mathrm{c}}^{\mathrm{PV}}$; one solves for $u_e = \Sigma w_e$ on the even sites, completes u on the odd sites as in Eq. (6.1), and sets $w = \Sigma u$.

C. Levels below the coarse lattice

Both factors may be preconditioned by further levels. The coarse sites are grouped into super-blocks; on each super-block, local vectors are formed from the coarse test vectors $\mathcal{P}^\dagger s_j$ and $\mathcal{P}^\dagger \hat{t}_j$, split by Σ and orthonormalized (Gram–Schmidt applied twice); the level operators are obtained by probing the nearest-neighbor stencil; the last level is inverted densely. With Q_ℓ the prolongation into level $\ell - 1$ and B_ℓ the diagonal blocks of level ℓ:

Quark factor. A V-cycle with the coarse correction followed by one block-Jacobi sweep on the updated residual, without pre-smoothing,

$$z = Q_{\ell+1} M_{\ell+1} Q_{\ell+1}^\dagger r, \qquad M_\ell r = z + B_\ell^{-1}(r - \mathcal{D}_\ell z); \tag{6.3}$$

on the coarse lattice itself the same multiplicative combination is used, and with the even-odd coarse form its even part preconditions the Schur system. The level test vectors can optionally be refined by a few GCR steps preconditioned by the levels already built.

Pauli–Villars factor (Σ-Hermitian form only). The additive Hermitian positive preconditioner

$$M_\ell = |B_\ell|^{-1} + Q_{\ell+1} M_{\ell+1} Q_{\ell+1}^\dagger, \qquad M_{\ell_{\max}} = |\mathcal{H}_{\ell_{\max}}|^{-1}, \tag{6.4}$$

in the spirit of absolute-value preconditioning [25].

VII. Setup

The setup produces the vectors $y_1, \dots, y_n$ on the even sites, then the basis $\{s_j, \hat{t}_j\}$, $\mathcal{P}$, and the coarse operators. It always uses the right form; the same setup serves the left forms and every α (Sec. V).

A. Chebyshev-filtered subspace iteration

The vectors approximate the right singular vectors of the smallest singular values of $\tilde{A}_R$, i.e. eigenvectors of the Hermitian positive operator $H_n = \tilde{A}_R^\dagger \tilde{A}_R$. With the Chebyshev polynomial T_d and $\ell(x) = \big(2x - (\lambda_{\mathrm{hi}} + \lambda_{\mathrm{lo}})\big)/(\lambda_{\mathrm{hi}} - \lambda_{\mathrm{lo}})$, $|T_d(\ell(x))| \le 1$ on $[\lambda_{\mathrm{lo}}, \lambda_{\mathrm{hi}}]$ while components below λ_{lo} grow; $T_d(\ell(H_n))$ is applied by the three-term recurrence.

Algorithm 1. LOWMODES: Chebyshev-filtered subspace iteration on $H_n = \tilde{A}_R^\dagger \tilde{A}_R$.

1: $X \leftarrow n$ random vectors on the even sites
2: $\lambda_{\mathrm{hi}} \leftarrow$ a bound slightly above $\lambda_{\max}(H_n)$ (power iteration); $\lambda_{\mathrm{lo}} \leftarrow$ a small fraction of λ_{hi}
3: **for** each round **do**
4: $d \leftarrow$ a degree proportional to $\sqrt{\lambda_{\mathrm{hi}}/\lambda_{\mathrm{lo}}}$, within fixed bounds
5: $X \leftarrow T_d(\ell(H_n))\, X$; orthonormalize X through its Gram matrix
6: $G \leftarrow X^\dagger H_n X = V \Theta V^\dagger$; $X \leftarrow XV$ ▷ Rayleigh–Ritz
7: $\lambda_{\mathrm{lo}} \leftarrow \theta_{\max}$ ▷ largest Ritz value
8: **end for**
9: **return** X, Θ

B. Bootstrap

Each bootstrap round builds the basis and the two-grid cycle of Sec. VIII from the current y, replaces every y_j by the result of a fixed small number of FGMRES steps on $\tilde{A}_R x = y_j$ from $x = 0$ preconditioned by that cycle, and orthonormalizes the new y through their Gram matrix ($y \leftarrow yV\Lambda^{-1/2}$ with $y^\dagger y = V\Lambda V^\dagger$). The cycle damps the components it already handles, so the new vectors are enriched in what it does not.

Algorithm 2. Setup of the two-grid method.

1: $y \leftarrow \mathrm{LOWMODES}(H_n)$
2: **for** each bootstrap round **do**
3: build $\mathcal{P}$ from $\{s_j, \hat{t}_j\}$ of y and the cycle $\mathcal{M}$ on it
4: **for** $j = 1, \dots, n$ **do**
5: $y_j \leftarrow$ a few FGMRES steps on $\tilde{A}_R x = y_j$, preconditioned by $\mathcal{M}$
6: **end for**
7: orthonormalize y
8: **end for**
9: compute $s_j, \hat{t}_j$ by Eqs. (5.4)–(5.5); split by Γ_5; orthonormalize per block $\to \mathcal{P}$
10: assemble $\mathcal{D}_{\mathrm{c}}$, $\mathcal{D}_{\mathrm{c}}^{\mathrm{PV}}$, the inverses of their diagonal blocks, their even-odd forms
11: optionally, build the levels below from $\mathcal{P}^\dagger s_j$, $\mathcal{P}^\dagger \hat{t}_j$

VIII. The Chebyshev smoother and the cycle

A. Smoother

The smoother is ν steps of Chebyshev iteration [26, 27] for $\tilde{A}x = b$ from $x = 0$, i.e. a fixed polynomial $x = p(\tilde{A})b$ of degree $\nu - 1$. It is defined by an ellipse with real foci $[\lambda_{\mathrm{lo}}, \lambda_{\mathrm{hi}}]$: λ_{hi} slightly above an estimate of the spectral radius of $\tilde{A}$ from a power iteration, and $\lambda_{\mathrm{lo}} = \lambda_{\mathrm{hi}}/\kappa_s$ for a chosen ratio κ_s. With $\theta = \frac{1}{2}(\lambda_{\mathrm{hi}} + \lambda_{\mathrm{lo}})$, $\delta = \frac{1}{2}(\lambda_{\mathrm{hi}} - \lambda_{\mathrm{lo}})$, $\sigma = \theta/\delta$, $\varrho_0 = 1/\sigma$, $r_0 = b$ and $d_0 = r_0/\theta$, step k is

$$x \leftarrow x + d_k, \quad r_{k+1} = r_k - \tilde{A}d_k, \quad \varrho_{k+1} = \frac{1}{2\sigma - \varrho_k}, \quad d_{k+1} = \varrho_{k+1}\varrho_k\, d_k + \frac{2\varrho_{k+1}}{\delta}\, r_{k+1}, \tag{8.1}$$

the last step only updating x; ν steps cost $\nu - 1$ applications of $\tilde{A}$. Only a real focal interval is used. The ratio κ_s is not derived from the operator and has to be chosen for it: the polynomial damps the part of the spectrum above λ_{lo} and leaves the rest to the coarse correction, and too wide or too narrow an interval can make the outer solve stagnate. The two setups of Sec. XI need $\kappa_s \approx 10$ ($M_5 = 1$, stout) and $\kappa_s \approx 30$ ($M_5 = 1.8$); the first stagnates with the second's interval, the second needs twice the iterations with the first's (Sec. H).

B. Cycle

Algorithm 3. Two-grid cycle $y = \mathcal{M}b$ for $\tilde{A}$.

1: $y \leftarrow p(\tilde{A})\, b$ ▷ pre-smoothing
2: $r \leftarrow b - \tilde{A}y$
3: $y \leftarrow y + e(r)$ ▷ correction (5.2)
4: $r \leftarrow b - \tilde{A}y$
5: $y \leftarrow y + p(\tilde{A})\, r$ ▷ post-smoothing

One cycle costs 2ν applications of $\tilde{A}$ (two smoothings and two residuals) plus the two coarse solves. Because the coarse solves are Krylov iterations stopped by a residual test, $\mathcal{M}$ is not a fixed linear map.

IX. The outer solver

The outer iteration solves $\tilde{A}y = \rho$ with the cycle as right preconditioner, and ψ is reconstructed as in Sec. IV. Since $\mathcal{M}$ varies from one application to the next, the outer method has to be flexible: FGMRES or flexible GCR, which keep the preconditioned directions; BiCGStab with the cycle as preconditioner is also available but has no flexible counterpart of its convergence theory. Flexibility alone gives no convergence guarantee. For a fixed $\mathcal{M}$, restarted or truncated methods converge for every restart length if the field of values of $\tilde{A}\mathcal{M}$ lies in the right half-plane [21]; otherwise only the full method is covered, by the spectrum and the pseudospectra of $\tilde{A}\mathcal{M}$ [13, 29], and a truncated one can stagnate [30]. For the present cycle the field of values is not in the right half-plane, the pseudospectra do bound the full method, and GCR(2) converges only because the coarse solves are inexact (Secs. C–F). Block variants treat several right-hand sides together. Convergence is monitored on the even-odd system; the residual of the original equation, $\|D\psi - \eta\|/\|\eta\|$, is evaluated with the plain D after reconstruction. The reference solver is CG on the normal equation of the even-odd operator $C_e^{-1}S$ (CGNE), which is robust but squares the condition number; it runs in double precision or in mixed precision, by defect correction in double precision with CG in single precision, each inner solve stopped where the outer tolerance is met.

X. Implementation in RutticeQCD

The method is implemented in RutticeQCD (Rust; branch `cuda-bridge`), on the CPU and on NVIDIA GPUs through CUDA kernels of the Wilson and Möbius operators taken from Bridge++. On the

TABLE 2. The two setups and the multigrid of the mass scan. All runs: Möbius $(b, c) = (1.5, 0.5)$, $L_s = 8$, $m_{\mathrm{PV}} = 1$, $\alpha_{\mathrm{PV}} = 1$, boundary $(1, 1, 1, -1)$, $\alpha = 0.6$ unless stated. Configurations: the v2 quenched test configurations on 12^4, 16^4 and $24^3 \times 48$ (the 12^4 one has one unpaired near-zero mode, $Q = +1$).

	setup S (stout)	setup B (bare)
M_5	1.0	1.8
gauge smearing	stout, $\rho = 0.1$, 3 steps	none
Chebyshev smoother ν, κ_s	8, 10	12, 30
multigrid (both setups)		
aggregates	4^4 ($24^3 \times 48$: levels below with super-blocks 2, 3)	
test vectors	32 (16 filtered low modes of $\tilde{A}_R^\dagger \tilde{A}_R$, lifted to $\{s, \hat{t}\}$)	
setup	filter 2 rounds; bootstrap 2 rounds of 4 FGMRES steps	
coarse solves	single precision, $\lvert r\rvert^2/\lvert b\rvert^2 < 10^{-2}$	
outer solver	GCR(2) on $\tilde{A}_L$ to $\lvert r\rvert^2/\lvert b\rvert^2 < 10^{-18}$	

device the fine operators, the prolongation (stored in half precision unless asked otherwise), the coarse operators and the coarse Krylov solvers all stay on the GPU. The cycle above the coarse lattice runs in single precision on operators converted on the device from the links of the double-precision ones; the outer solver is in double precision. The coarse solvers are BiCGStab for the quark factor and MINRES on $\mathcal{H}_{\mathrm{c}}$ for the Pauli–Villars factor (Sec. VI), both stopped at $|r|^2/|b|^2 < 10^{-2}$ unless stated, on the even coarse sites. The numbers below come from the parameter-file program `ruttice` and from the examples `preconditioned_fov` (CPU) and `preconditioned_fov_device` (GPU), whose options are given with each result. The eigensolvers (thick-restart Lanczos, Krylov–Schur Arnoldi [28] with left eigenvectors, and $\sigma_{\min}(z - X)$ by Lanczos on $(z - X)^\dagger(z - X)$) are in the module `eigen` and are checked against Bridge++'s eigensolver test.

XI. Numerical results

A. Setups

Timings are from an NVIDIA A30; an RTX 3080 was used for iteration counts and spectral quantities only. Table 2 lists the setups. The smoother interval is not chosen from the operator: setup S needs $\kappa_s \approx 10$ and setup B $\kappa_s \approx 30$; with B's interval setup S stagnates, with S's setup B needs twice the iterations (Sec. H).

B. Comparison with CGNE

Table 3 and Fig. 4 are the mass scan on $24^3 \times 48$. The CGNE iterations grow by a factor of 21 (S) and 18 (B) from $m = 0.1$ to $m = 0.001$, the multigrid iterations by 2 (S) and 3.7 (B). At $m = 0.001$ the multigrid solve is 4.7 times (S) and 6.0 times (B) faster; with the setup of about 390 s it pays for itself after 4 (S) and 3.4 (B) right-hand sides, i.e. within one point-source propagator (12 columns) for $m \leq 0.005$ (S) and $m \leq 0.01$ (B). For $m \geq 0.05$ in setup S the multigrid is not faster at all. The multigrid needs three times the device memory of CGNE; on the 24 GB A30 this excludes GCR(k) with $k \geq 8$ (out of memory after the setup). Neff's $\alpha = 0.6$ against $\alpha = 1$ in the left form: equal for S, one to three iterations fewer for B. Runs of the same parameter file with two builds (`535c9ef`, `ca1e892`) differ by up to two iterations in three of eight cases; the cause (code change or non-determinism of the device reductions) is not established.

Mixed-precision CGNE. For a comparison on equal terms the reference solver also runs in mixed precision (defect correction in double precision, CG in single precision on the normal equation of $C_e^{-1}S$, each

TABLE 3. Mass scan on $24^3 \times 48$, A30, one random right-hand side to $|r|^2/|b|^2 < 10^{-18}$ on the even-odd system: even-odd CGNE in double precision against the multigrid (GCR(2) on $\tilde{A}_L$), $\alpha = 0.6$ for both and $\alpha = 1$ for the multigrid; iterations, solve time in seconds, coarse-operator applications, and N_{rhs}, the number of right-hand sides at which the multigrid setup (376–394 s) is paid back, $t_{\mathrm{setup}}/(t_{\mathrm{CGNE}} - t_{\mathrm{MG}})$. Final $|D\psi - \eta|^2/|\eta|^2$: 10^{-17}–3×10^{-16} (CGNE), 2×10^{-19}–3×10^{-18} (MG). Device memory peak: CGNE 6.9 GB, MG 19.9 GB ($\alpha = 0.6$). RutticeQCD `535c9ef`.

	CGNE		MG, $\alpha = 0.6$			MG, $\alpha = 1$		
m	it	s	it	s	coarse	it	s	N_{rhs}
setup S ($M_5 = 1.0$, stout)								
0.1	173	5.7	13	11.2	1160	15	12.1	—
0.05	291	9.5	13	9.5	734	15	11.0	—
0.02	644	21.0	14	10.4	810	14	10.5	36
0.01	1166	37.9	15	11.7	1006	15	11.9	15
0.005	2067	67.1	18	13.7	1134	18	14.8	7.2
0.002	3419	110.9	23	19.6	1942	22	20.3	4.3
0.001	3602	116.8	26	24.9	2875	27	27.1	4.3
setup B ($M_5 = 1.8$)								
0.1	231	7.6	7	5.5	190	9	6.9	184
0.05	402	13.1	10	8.1	315	12	9.5	78
0.02	857	27.8	13	10.5	428	15	12.2	22
0.01	1447	46.9	15	12.5	568	17	14.1	11
0.005	2299	74.5	18	15.1	722	21	17.9	6.6
0.002	3473	112.5	23	19.8	1056	25	21.5	4.2
0.001	4228	137.0	26	22.8	1278	27	23.4	3.4

inner solve stopped where the outer tolerance is met; RutticeQCD `49bcf37`). It needs 1.0–1.2 times the iterations of double precision and 0.65–0.75 times the time, with the same final residuals (2×10^{-18}–2×10^{-16}) and 10.1 GB of device memory (Fig. 4; e.g. $m = 0.001$: 4017 iterations, 89.3 s (S); 4334, 96.4 s (B)). Against it the multigrid solve at $m = 0.001$ is 3.6 (S) and 4.2 (B) times faster, and the setup is paid back after 6 (S) and 5 (B) right-hand sides; for $m \geq 0.05$ (S) the mixed-precision CGNE solve is the faster one, and at $m = 0.1$ (B) the two take the same time.

C. Field of values

The Hermitian part of $\tilde{A}_R\mathcal{M}$ was computed with $\mathcal{M}$ made a fixed linear map: both coarse factors solved to $|r|^2/|b|^2 < 10^{-20}$ in double precision with the exact adjoints, and $\mathcal{M}^\dagger$ built with the same $\mathcal{P}$ and the same smoother interval, so that $\langle z, \tilde{A}_R\mathcal{M}w\rangle$ and $\langle \mathcal{M}^\dagger\tilde{A}_R^\dagger z, w\rangle$ agree to 10^{-10}. The cycle was also taken apart: the smoother alone, $S = p(\tilde{A}_R)$, the coarse correction alone, E of Eq. (5.2), and the cycle $\mathcal{M}$ (Table 4).

On $4^3 \times 8$ the field of values of $\tilde{A}_R\mathcal{M}$ lies in the right half-plane; on 12^4 and 16^4 it does not. The smoother is harmless ($\|I - \tilde{A}_R S\| \approx 1$); the factor that amplifies is the coarse correction, $\|I - \tilde{A}_R E\|$ of order 10^3 to 10^4 against 18.5 on $4^3 \times 8$: $\tilde{A}_R E$ is the identity on the vectors the correction is built from (Proposition 1) but amplifies other directions by three to four orders of magnitude, like a strongly oblique projection. The bound of Ref. [21] for GCR therefore does not apply (open problem 5). Replacing E by the Galerkin correction of $\tilde{A}_R$ itself, $\mathcal{P}(\mathcal{P}^\dagger\tilde{A}_R\mathcal{P})^{-1}\mathcal{P}^\dagger$ (built densely), brings the field of values of setup B into the right half-plane, $[0.121, 1.99]$, and reduces $\|I - \tilde{A}_R\mathcal{M}\|$ to 1.84 (setup S: $[-3.62, 13.1]$,

TABLE 4. Real part of the field of values $W(X)$ and $\|I - X\|_2$ for $X = \tilde{A}_R S$, $\tilde{A}_R E$, $\tilde{A}_R \mathcal{M}$ (exact coarse solves). $m = 0.05$ on $4^3 \times 8$ (2^4 aggregates, 8 filtered low modes), $m = 0.001$ otherwise, with the multigrid of the mass scan (16 low modes, 32 test vectors, 4^4, Chebyshev of Table 2) but no levels below the coarse lattice. `preconditioned_fov_device -parts`.

	$\operatorname{Re} W(\tilde{A}_R)$	$\|I - \tilde{A}_R S\|$	$\|I - \tilde{A}_R E\|$	$\operatorname{Re} W(\tilde{A}_R \mathcal{M})$	$\|I - \tilde{A}_R \mathcal{M}\|$
$4^3 \times 8$, $M_5 = 1$	$[-5.0 \times 10^{-3}, 2.26]$		18.5	$[0.772, 1.222]$	
12^4 S	$[-4.59 \times 10^{-3}, 1.98]$	1.018	1.94×10^4	$[-23.9, 30.8]$	54.8
12^4 B	$[-5.02 \times 10^{-2}, 4.46]$	1.112	2.86×10^3	$[-1.49, 4.17]$	5.64
16^4 S			1.54×10^4	$[-15.6, 27.1]$	
16^4 B			2.5×10^3	$[-1.42, 4.27]$	

TABLE 5. Eigenvalues of $\tilde{A}\mathcal{M}$ (exact coarse solves) by Krylov–Schur: the smallest real part, the condition numbers $\kappa = 1/|y^\dagger x|$ of the six eigenvalues of smallest real part (left eigenvectors from $\mathcal{M}^\dagger \tilde{A}^\dagger$), the eigenvalues farthest from 1 with their κ, and the radius about 1 of the disc that holds all the others. $m = 0.001$. `preconditioned_fov_device -arnoldi -far 1,0` (`-left` for $\tilde{A}_L$).

	$\min \operatorname{Re} \lambda$	κ (6 lowest)	outliers (κ)	others: $\|\lambda - 1\| \le$
$4^3 \times 8$, $m = 0.05$, R	0.857	2–4		
$4^3 \times 8$, $m = 0.05$, L	0.854	2.8		0.156 (Ritz)
12^4 S, R	0.868	6.3–14.5	$5.86 + 0.86i$ (11.2)	0.434
12^4 S, L	0.861	7.2–26.2	$5.40 + 0.89i$ (13.9)	0.455
12^4 B, R	0.831	9.4–20.3		0.753
12^4 B, L	0.830			0.747
16^4 S, R	0.794	6.2–12.8	$10.63 + 0.19i$ (4.38)	0.474
			$3.13 + 0.02i$ (4.74)	
16^4 B, R	0.783	13.5–42.9		0.840

16.8); but GCR(8) to 10^{-16} then needs more iterations, 16 against 13 (B) and 12 against 10 (S). The field of values is not what decides the iteration count.

D. Spectrum

Although the field of values reaches far into the left half-plane, every eigenvalue lies at $\operatorname{Re} \lambda \ge 0.78$ (Table 5), with moderate condition numbers. Setup S has one (12^4) or two (16^4) isolated outliers of large modulus that grow with the volume; all other eigenvalues lie within 0.43–0.47 of 1. The left and right forms have the same spectrum to the precision shown. At $m = 0.005$ on 16^4 the lowest real parts are 0.802 (S) and 0.790 (B).

E. Pseudospectra and a bound for GMRES

For a non-normal operator the eigenvalues alone do not bound the convergence of GMRES; the ε-pseudospectrum, the set where $\sigma_{\min}(z - X) < \varepsilon$, does [13, 29]. If a contour Γ of length L encloses the spectrum of X but not the origin and $\sigma_{\min}(z - X) \ge \varepsilon$ on it, then for any polynomial p_k with $p_k(0) = 1$,

$$\frac{\|r_k\|}{\|r_0\|} \le \|p_k(X)\| \le \frac{L}{2\pi\varepsilon} \max_{z \in \Gamma} |p_k(z)|. \tag{11.1}$$

An isolated outlier λ_o outside Γ is removed by the factor $(1 - z/\lambda_o)$ of p_k: then $p_k(X) = p_k(X)(I - P_o)$ with P_o the spectral projector, $\|(z - X)^{-1}(I - P_o)\| \le \kappa_o/\varepsilon$, and with a circle $|z - c| = r$ and

TABLE 6. $\min_\Gamma \sigma_{\min}(z - \tilde{A}_R\mathcal{M})$ on circles $|z-c| = r$ (16 points each), and the bound (11.1) or (11.2): the factor per iteration r/c, the constant, and the iterations it guarantees for $\|r_k\|/\|r_0\| < 10^{-9}$. Lanczos to 10^{-3} relative in σ^2. `preconditioned_fov_device -pseudo`.

	c	r	$\min \sigma_{\min}$	r/c	constant	k for 10^{-9}
12^4 S	1	0.6	0.0285	0.6	220	53
12^4 S	1	0.5	0.0107	0.5	480	40
12^4 B	1.4	0.93	0.0985	0.66	9.4	57
12^4 B	1.3	0.87	0.0860	0.67	10.1	58
16^4 S	1	0.6	0.0262	0.6	190	53
16^4 S	1	0.55	0.0135	0.55	340	47
16^4 B	1.4	1.05	0.1295	0.75	8.1	80

$p_k(z) = (1 - z/\lambda_o)(1 - z/c)^{k-1}$,

$$\frac{\|r_k\|}{\|r_0\|} \le \kappa_o \frac{r}{\varepsilon} \frac{|\lambda_o - c| + r}{|\lambda_o|} \left(\frac{r}{c}\right)^{k-1}. \tag{11.2}$$

On 16^4 setup S has two outliers; the factor κ_o is then replaced by $\|I - P_1 - P_2\| \le 1 + \kappa_1 + \kappa_2$ and there are two factors of the form $(|\lambda_o - c| + r)/|\lambda_o|$. On the real axis between the origin and the spectrum, $\sigma_{\min}(z - \tilde{A}_R\mathcal{M}) \ge 0.024$ (S) and ≥ 0.031 (B) on 12^4 ($z = -0.5 \ldots 0.7$), and 0.107 (S), 0.271 (B) at the origin.

The bound guarantees convergence at a fixed rate, r/c per iteration, without stagnation; it is pessimistic by a factor of 3–5 in iterations (GCR needs 10–17 with exact coarse solves, Sec. F), and its rate is that measured on $24^3 \times 48$, where the median factor per GCR step of $|r|^2$ is 0.23–0.34 (Table 8), i.e. 0.48–0.58 in $|r|$, against $r/c = 0.5$–0.6 for setup S. Setup B is not worse between 12^4 and 16^4 ($\min \sigma$ grows from 0.098 to 0.13); in setup S the outliers grow but their condition numbers fall and ε stays the same.

What is not yet rigorous. (i) $\sigma_{\min}$ is known at 16 points per circle, spaced by more than ε; since $\sigma_{\min}(z - X)$ is 1-Lipschitz in z, a lower bound on the whole circle needs a spacing below ε. [TODO: the dense contours on 12^4 (160 and 96 points), running.] (ii) Lanczos approaches $\sigma^2_{\min}$ from above; a rigorous lower bound needs an inclusion theorem (Kato–Temple) with a bound on the second singular value. (iii) That the disc holds all eigenvalues but the outliers rests on the four eigenvalues farthest from c found by Arnoldi. (iv) The bound is for full GMRES with a fixed linear $\mathcal{M}$ (Sec. F). (v) The FOV and spectra are not available on $24^3 \times 48$: the eigensolvers do not fit in the memory of the A30.

F. Truncated GCR

With the usual coarse solves to 10^{-2}, GCR(2) needs one to five iterations more than the full method (Table 7). With accurate coarse solves, where $\mathcal{M}$ is close to the fixed map of Secs. C–E, GCR(2) stagnates ($|r|^2 \approx 3 \times 10^{-8}$ after 1000 iterations) while GCR(8) converges in 26 and GCR(≥ 16) in 17. This is what the analysis predicts: the pseudospectra guarantee the full method; a truncated or restarted one has no such guarantee when the field of values contains the origin [21, 30]. That the truncated solver works with inexact coarse solves is an empirical fact; a plausible reason, not shown here, is that a coarse Krylov solve stopped at 10^{-2} does not resolve the directions in which the exact correction E amplifies. On $24^3 \times 48$ the full method cannot be run on the A30: GCR(8) and GCR(16) run out of device memory after the setup (GCR(2) peaks at 19.9 of 24 GB, and every further direction is another fine-grid vector).

No run shows a plateau (Table 8, Fig. 6); the first step reduces $|r|^2$ by about 10^{-5} in every run.

TABLE 7. GCR(k) iterations to $|r|^2/|b|^2 < 10^{-18}$, 12^4, $m = 0.001$, the multigrid of Table 2 (no levels below the coarse lattice), one random right-hand side; the point source: range over its 12 columns.

	coarse solves	$k = 2$	8	16	32	64
S, $\tilde{A}_L$	10^{-2}	14			12	
S, $\tilde{A}_R$	10^{-2}	12			11	
B, $\tilde{A}_L$	10^{-2}	14			13	
B, $\tilde{A}_R$	10^{-2}	15			14	
S, $\tilde{A}_L$, point source	10^{-2}	15–19			13–14	
S, $\tilde{A}_L$	10^{-8}	no conv.	26	17	17	17
S, $\tilde{A}_L$	10^{-4}	18			16	

TABLE 8. Residual history of GCR(2), $24^3 \times 48$, $\tilde{A}_L$, the multigrid of Table 2: steps, median and largest factor of $|r|^2$ per step, and the longest stretch of steps with a factor above $1/2$.

setup	m, α	steps	median	worst	slow stretch
S	0.001, 0.6	28	0.315	0.564	2 steps
S	0.001, 1.0	27	0.338	0.479	none
S	0.002, 0.6	23	0.233	0.426	none
S	0.002, 1.0	23	0.253	0.354	none
B	0.001, 0.6	26	0.299	0.357	none
B	0.001, 1.0	28	0.317	0.403	none
B	0.002, 0.6	23	0.246	0.300	none
B	0.002, 1.0	25	0.280	0.347	none

G. Γ_5-Hermiticity for $c \neq 0$

For $c = 0$ the coarse operators are Σ-Hermitian to rounding; for $c \neq 0$ they are not, by a few percent in random directions and up to 14% in the directions of the smallest singular values, growing linearly with c (Table 9), as Eq. (3.4) says. This does not hurt the method: with MINRES on $\mathcal{H}_c$ for the Pauli–Villars factor the outer iterations at $c = 0.5$ are those of the Shamir operator ($c = 0$, a different operator, for which MINRES is exact), and MINRES is better than BiCGStab on the exact adjoint in every case (Table 10); with BiCGStab the stout setup nearly stagnates. The non-Hermitian part of $\mathcal{H}_c$ is small enough for MINRES stopped at 10^{-2}; why BiCGStab is so much worse, also at $c = 0$, has not been studied.

H. Sensitivity to the parameters

The method is robust within a window of each parameter and fails outside it (Table 11). The smoother interval must suit the setup; the bootstrap is necessary; more accurate coarse solves are not better, and with GCR(2) accurate ones stagnate (Sec. F). Larger setups (smaller aggregates, more test vectors) buy fewer iterations at a setup cost that only many right-hand sides repay. All runs of the table use GCR(2); the stagnations are of the kind of Sec. F, and whether full GCR would remove them has not been tested. The table gives iterations only (RTX 3080); its timings on the A30, and Lüscher's test of local coherence for these test vectors, have not been measured.

TABLE 9. $\delta(v) = \|\Sigma X \Sigma v - X^\dagger v\| / \|X^\dagger v\|$ for the coarse operators $X = \mathcal{D}_c = \mathcal{P}^\dagger D \mathcal{P}$ and $\mathcal{P}^\dagger D_{\rm PV} \mathcal{P}$, $X^\dagger$ the exact adjoint, Σ the coarse Γ_5; $b = 1 + c$, $L_s = 8$, $m = 0.001$ ($4^3 \times 8$: $m = 0.05$, 2^4 aggregates, 32 coarse components; 12^4: 4^4, 64). Random v (four) and the right singular vectors of the four smallest singular values. `examples/coarse_hermiticity`. *Lanczos on $X^\dagger X$ not converged in 2000 restarts (the low end of $\mathcal{D}_c$ is crowded): approximate vectors.

	$\mathcal{P}^\dagger D \mathcal{P}$		$\mathcal{P}^\dagger D_{\rm PV} \mathcal{P}$	
	random	low	random	low
$4^3 \times 8$, $M_5 = 1$, $c = 0$	3.8×10^{-16}	9×10^{-16}	4.0×10^{-16}	9×10^{-16}
$4^3 \times 8$, $M_5 = 1$, $c = 0.5$	0.041	0.065–0.074	0.050	0.090–0.095
12^4 S, $c = 0.25$	0.012	0.034–0.050*	0.030	0.023–0.031
12^4 S, $c = 0.5$	0.023	0.067–0.099*	0.053	0.041–0.054
12^4 B, $c = 0.5$	0.030	0.140–0.146	0.055	0.053–0.066

TABLE 10. GCR(2) iterations (coarse applications) with the Pauli–Villars factor solved by MINRES on $\mathcal{H}_c$, which assumes $\delta = 0$, or by BiCGStab on the exact adjoint, both to 10^{-2}; 12^4, $m = 0.001$, $\tilde{A}_L$, otherwise Table 2.

	$c = 0.5$, $b = 1.5$		$c = 0$, $b = 1$	
	MINRES	BiCGStab	MINRES	BiCGStab
setup S	14 (3098)	373 (192892)	13 (3061)	24 (7276)
setup B	14 (2351)	19 (3876)	15 (2103)	28 (3540)

XII. Reproducing and extending the measurements

All numbers of Sec. XI come from one configuration per volume; the most useful extensions are more configurations, a larger volume and dynamical ensembles. This section gives what is needed to run them.

Code. RutticeQCD, `https://github.com/atsanroger/RutticeQCD`, branch `cuda-bridge` (the commit of each result is given with it; later commits give the same iterations, up to the two-iteration differences between builds noted in Sec. B). It needs Rust (stable) and, for the GPU, CUDA 12:

```
export CUDA_HOME=/usr/local/cuda
cargo build --release --features cuda-bridge
./target/release/ruttice mg_24.yaml
```

The GPU backend runs on one GPU. The parameter files are in the style of Bridge++ and are documented in `docs/USAGE.md`; gauge configurations are read in the Bridge++ text and binary formats, NERSC, ILDG and SciDAC.

The mass scan (Table 3, Fig. 4). The parameter files are `docs/mg_background/runs/mg_24.yaml` (the multigrid) and `runs/cgne_24.yaml` (the reference, double or mixed precision); the comments in them say how to switch to setup B and to a point source. Each run prints a line

```
seed 91: GCR 26 iterations on the device; 2875 coarse applications;
    worst |Dx-b|^2 2e-18; 24.9 s; ... device 19.91 GB
```

and, for the multigrid, `setup on the device: 393 s`. These, per mass and setup, are what Table 3 lists. On an A30 one configuration takes about 1.5 hours for the multigrid (seven masses, both setups; the setup dominates) and about 40 minutes for the two CGNE scans.

TABLE 11. GCR(2) iterations when one parameter of Table 2 is changed, 12^4, $m = 0.001$, $\tilde{A}_L$, one random right-hand side (base: 14 in both setups); "stag.": not converged in 1000 iterations. Setup times in seconds where they change much.

change	setup S	setup B
aggregates 2^4 / 3^4 / 6^4	9 (89 s) / 12 / 23	9 (61 s) / 11 / 20
test vectors 8 / 16 / 24 / 48	stag. / 27 / 22 / 10 (40 s)	39 / 19 / 16 / 11 (30 s)
smoother $\nu = 4$ / 6 / 8 / 12 / 16	31 / 18 / 14 / 12 / 9	stag. / 84 / 26 / 14 / 13
interval $\kappa_s = 5$ / 10 / 30 / 100	22 / 14 / stag. / stag.	stag. / 31 / 14 / stag.
coarse solves 10^{-1} / 10^{-2} / 10^{-4} / 10^{-8}	23 / 14 / 18 / stag.	19 / 14 / 16 / 17
bootstrap rounds 0 / 1 / 2 / 3	stag. / 16 / 14 / 23	249 / 47 / 14 / 12
right form $\tilde{A}_R$	12	15

Memory and volume. On $24^3 \times 48$ the multigrid needs 19.9 GB of device memory (CGNE 6.9 GB in double, 10.1 GB mixed), roughly proportional to the volume: $32^3 \times 64$ needs about 63 GB (an 80 GB GPU), $48^3 \times 96$ more than one GPU has. The aggregates, 4^4, must divide the lattice, and each level below the coarse lattice must divide the extent of the level above (`MultiGrid_Level2`, `_Level3`; on $32^3 \times 64$, for instance, super-blocks 2 and 2).

The spectral quantities (Secs. C–E). The example `preconditioned_fov_device` computes, with exact coarse solves, the field of values, the parts of the cycle, the eigenvalues with their condition numbers, and $\sigma_{\min}(z - \tilde{A}\mathcal{M})$ on contours, e.g.

```
cargo run --release --features cuda-bridge \
    --example preconditioned_fov_device -- \
    16 16 binary conf.bin 1.0 0.001 1 --ratio 10 \
    --arnoldi --far 1,0 --pseudo --contour 1,0,0.6,16
```

(arguments: L, T, format, file, M_5, m, stout 0/1). It fits 16^4 in 24 GB and takes about an hour there; $24^3 \times 48$ does not fit. The CPU examples `preconditioned_fov` and `coarse_hermiticity` need no GPU.

What would help most.

1. The mass scan of Table 3 on three to five further $24^3 \times 48$ configurations, for the spread of the iterations and of the speed-up.
2. The same on a larger volume ($32^3 \times 64$ on an 80 GB GPU), for the volume dependence of the multigrid iterations and of the setup.
3. The mass scan on a dynamical ($2 + 1$-flavor) ensemble, e.g. that of Ref. [22], where the spectra of Sec. II were taken.
4. The spectral run above on further 16^4 configurations, for how the outliers of setup S and the pseudospectra vary between configurations.

For each run, the result lines above and the commit of the build are enough; the logs contain everything else.

XIII. Open problems

The following are questions of theory and algorithm, not findings.

1. **Exactness for the left form.** Proposition 1 holds for the right form. For $\tilde{A}_L$ (with or without α) the Pauli–Villars lift that would be needed differs from $\hat{t}$ by site-local factors (second remark of Sec. C). Whether a basis can be made exact for both forms at once, and what is lost by using the right-form basis for the left form, is open. Numerically the loss is small for the setups of Sec. XI: with the right-form basis, $\tilde{A}_L\mathcal{M}$ and $\tilde{A}_R\mathcal{M}$ have the same spectrum to three digits (Table 5) and the iterations differ by one to three (Table 7).

2. **Loss of Γ_5-Hermiticity for $c \neq 0$.** By Eq. (3.4), $\Gamma_5 D \Gamma_5 \neq D^\dagger$ at $O(c)$. MINRES on $\mathcal{H}_\mathrm{c} = \mathcal{D}_\mathrm{c}^\mathrm{PV}\Sigma$, the absolute-value block preconditioner (built from the Hermitian part only), the additive level preconditioner, and the Σ-adjoint in place of the exact adjoint are all exact only for $c = 0$. The exactly Hermitian operator $\Gamma_5 D_-^{-1} D$ [8] involves a non-local inverse. How large the non-Hermitian part is on the coarse space, and how to treat it, is open. On the coarse space it is a few percent, up to 14% in the lowest singular directions, linear in c; MINRES on $\mathcal{H}_\mathrm{c}$ is nevertheless as effective at $c = 0.5$ as at $c = 0$ (Sec. G). Its effect on the other components listed here has not been measured separately.

3. **Smoother interval for a non-normal operator.** The interval $[\lambda_\mathrm{hi}/\kappa_s, \lambda_\mathrm{hi}]$ is set from an estimate of the spectral radius, whereas the setup works with singular values (eigenvalues of $\tilde{A}_R^\dagger \tilde{A}_R$). For a non-normal $\tilde{A}$ these need not agree. Which part of the low end is left to the smoother after the coarse correction, and whether an ellipse with complex foci should replace the real interval, are open. That the choice matters is shown in Table 11: with GCR(2), a ratio three times too large stagnates in both setups, and too small ones cost up to twice the iterations or, six times too small, stagnate.

4. **Even-odd smoother and coarse space.** Boyle and Yamaguchi [18] found that a red-black smoother combined with the HDCR coarse correction was not effective, and attributed it to the eigenvectors of the red-black operator and of $\gamma_5 R_5 D$ not being aligned. In the present method the smoother and the correction both approximate $\tilde{A}^{-1}$ and the coarse space is built from low modes of $\tilde{A}_R$, so that particular mismatch is absent by construction; but the coarse operators are Galerkin projections of D and D_PV, not of $\tilde{A}$. Whether the even-odd outer system is preferable to the full-lattice A' for this method has not been established.

5. **Field of values versus real parts of eigenvalues.** Section II shows $\mathrm{Re}\,\lambda > 0$ for A'. The GCR bound [21] needs a positive definite Hermitian part, i.e. a field of values in the right half-plane, which is a stronger condition. Whether it holds for A', $\tilde{A}_R$ and $\tilde{A}_L$, and whether a shift of the kind used in Ref. [20] is needed, is not known. It fails slightly for $\tilde{A}_R$ itself and strongly for the preconditioned operator $\tilde{A}_R\mathcal{M}$ on 12^4 and 16^4 (Sec. C); the spectrum stays at $\mathrm{Re}\,\lambda \geq 0.78$ and the pseudospectra give a bound for full GMRES (Sec. E), but not for truncated GCR (Sec. F).

6. **Outer solver with a varying preconditioner.** Because the coarse solves are truncated, $\mathcal{M}$ is nonlinear and depends on the right-hand side. FGMRES and flexible GCR allow this; BiCGStab does not, and its behavior with $\mathcal{M}$ as preconditioner has no theoretical justification. The truncated GCR(2) used in practice converges with coarse solves to 10^{-2} and stagnates with accurate ones (Sec. F); why the inexact correction behaves better is open.

7. **Solving with $D^\dagger$.** The method is built for $D\psi = \eta$. A solve with $D^\dagger$, as needed for the normal equations in HMC, requires the dual correction (the two factors and the roles of the lifts exchanged), and its basis and exactness have not been derived. A cycle with D and D_PV exchanged on the same $\mathcal{P}$ is implemented; with exact coarse solves it is the adjoint of $\mathcal{M}$ (Sec. C), and RutticeQCD uses it for the solves with $\tilde{A}_R^\dagger$ in HMC. How good the basis built for D is for this dual correction has not been studied.

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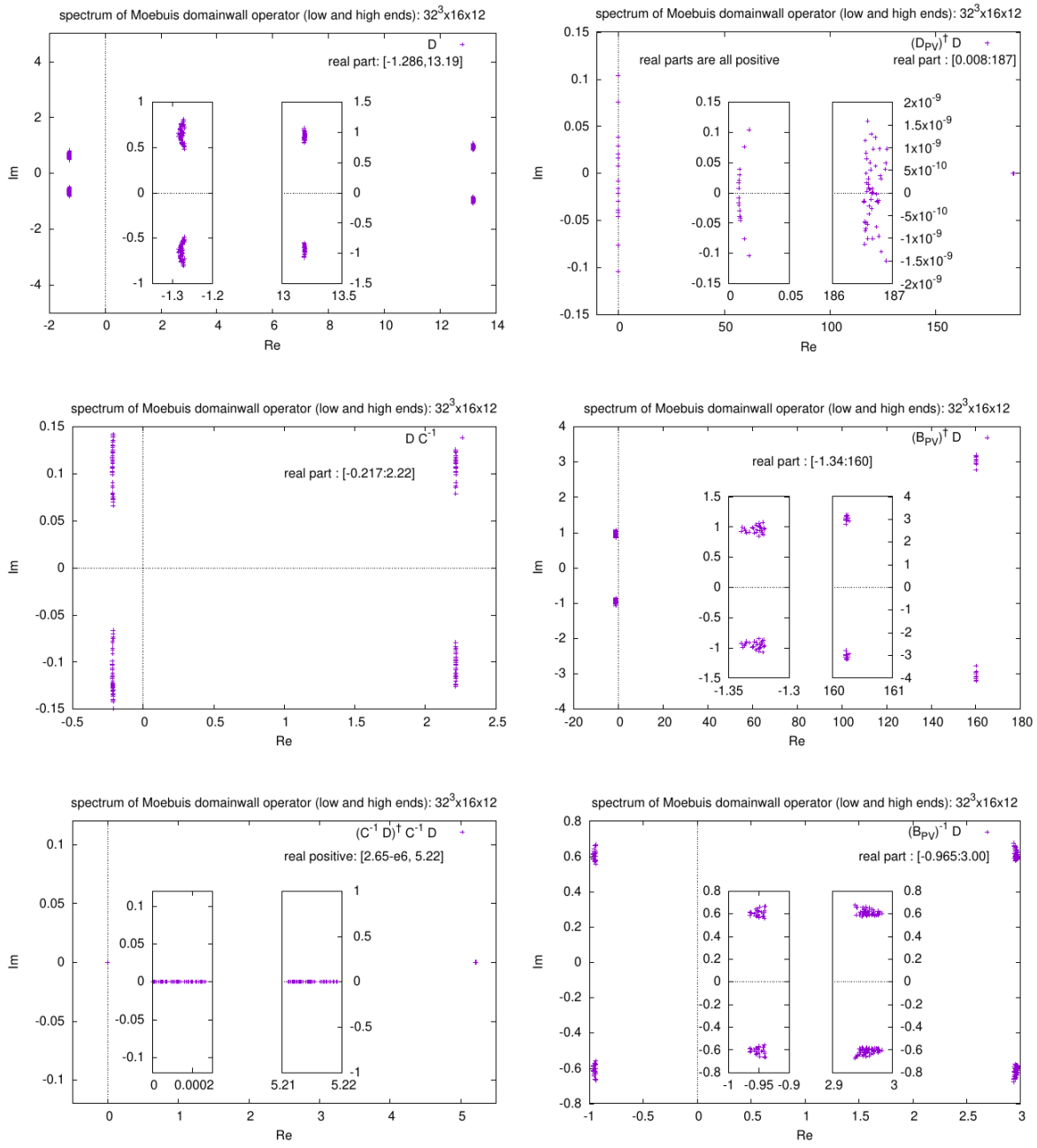


FIG. 2. Low and high ends of the spectrum of the Möbius operator on a JLQCD $2+1$-flavor configuration at the physical point, with preconditioners. Left column: D, DC^{-1}, $(C^{-1}D)^\dagger C^{-1}D$. Right column: $D_{\rm PV}^\dagger D$, $B_{\rm PV}^\dagger D$, $B_{\rm PV}^{-1}D$, where $B_{\rm PV}$ is the block-diagonal approximation of $D_{\rm PV}$ (inverted approximately in the last panel). The text of Ref. [22] quotes $48^3 \times 16$ with $L_s = 12$; the panel titles read $32^3 \times 16 \times 12$. Reproduced from Ref. [22], Fig. 3.

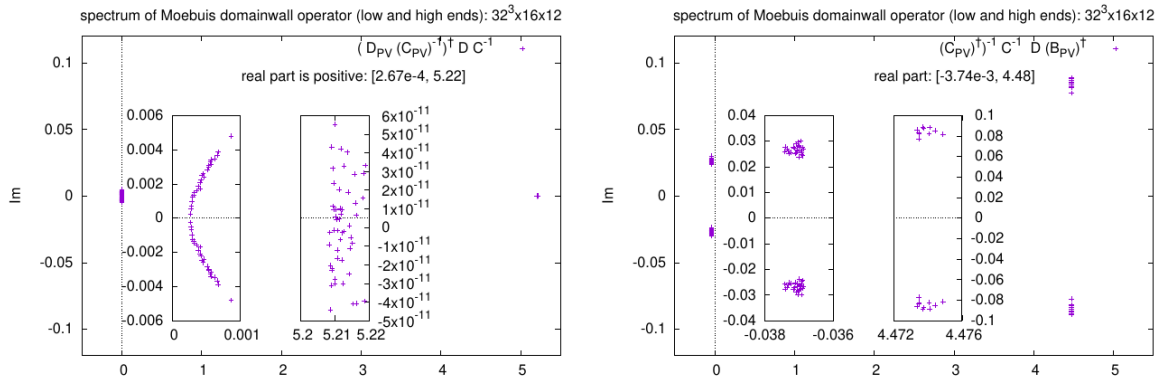


FIG. 3. Left: spectrum of $A' = (D_{\rm PV}C_{\rm PV}^{-1})^\dagger DC^{-1}$, Eq. (3.7), on the configuration of Fig. 2; all real parts are positive. Right: the same with $D_{\rm PV}$ replaced by its block-diagonal approximation $B_{\rm PV}$; negative real parts reappear. Lattice size as in Fig. 2 (the text of the source quotes $48^3 \times 16$, $L_s = 12$; the panel titles read $32^3 \times 16 \times 12$). Reproduced from Ref. [22], Fig. 4.

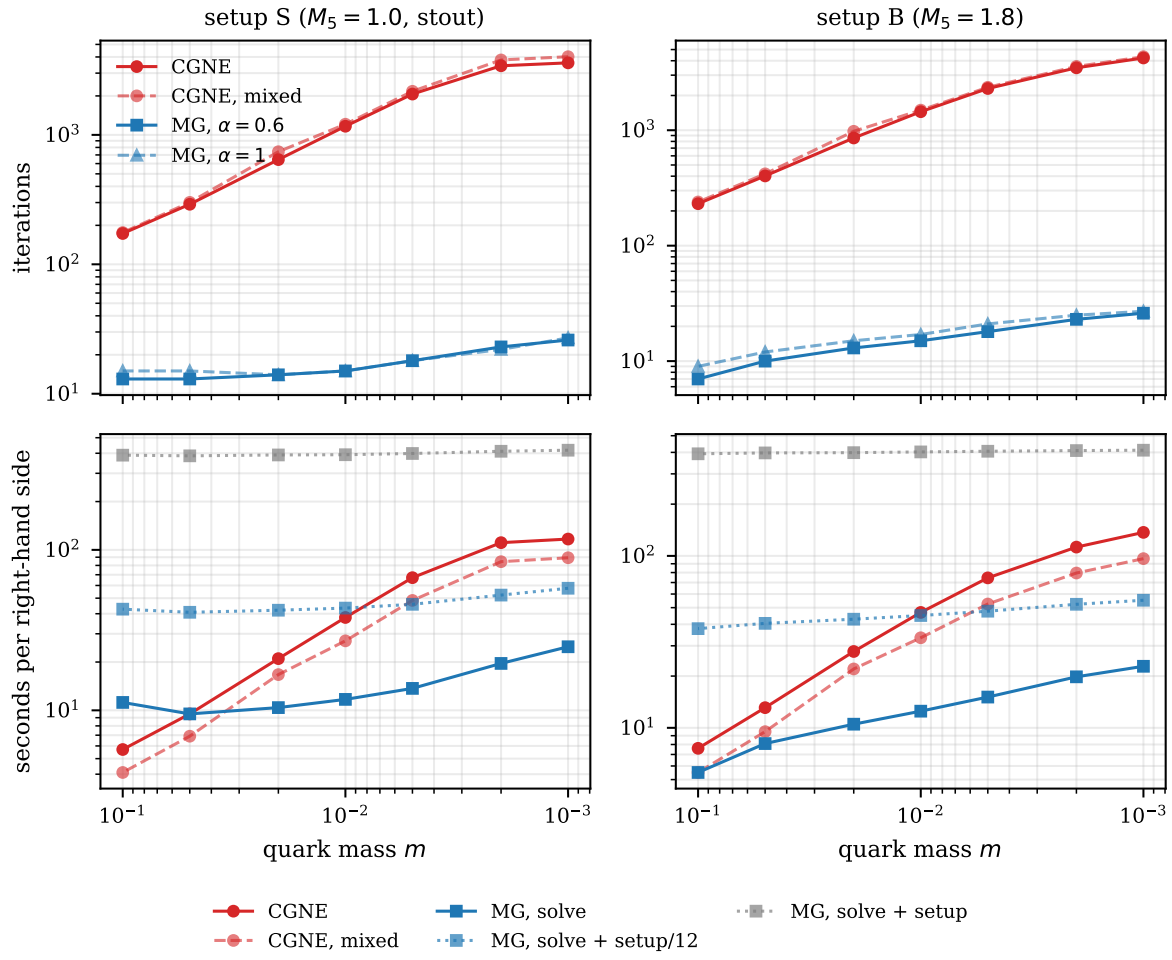


FIG. 4. The mass scan of Table 3 and the mixed-precision CGNE: iterations (top) and seconds per right-hand side (bottom) against the quark mass, $24^3 \times 48$, A30. "Solve + setup/12": the setup spread over the 12 columns of one point-source propagator. `data/massscan_24.tsv`, `figures.py`.

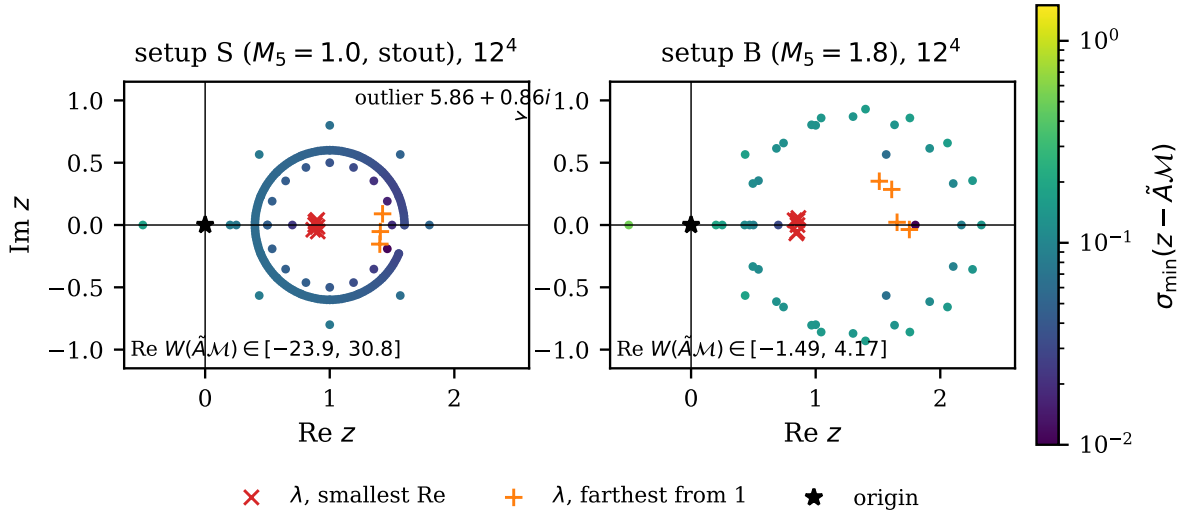


FIG. 5. 12^4, $m = 0.001$, exact coarse solves: the six eigenvalues of $\tilde{A}_R\mathcal{M}$ of smallest real part, the four farthest from 1, and $\sigma_{\min}(z - \tilde{A}_R\mathcal{M})$ (color) at the points where it was computed: the real axis, the contours of Table 6 and, for setup S, the dense circle $|z - 1| = 0.6$. The field of values reaches far into the left half-plane (its real range in the corner), the spectrum does not. `data/pseudo_*.tsv`, `data/spectrum_*.tsv`.

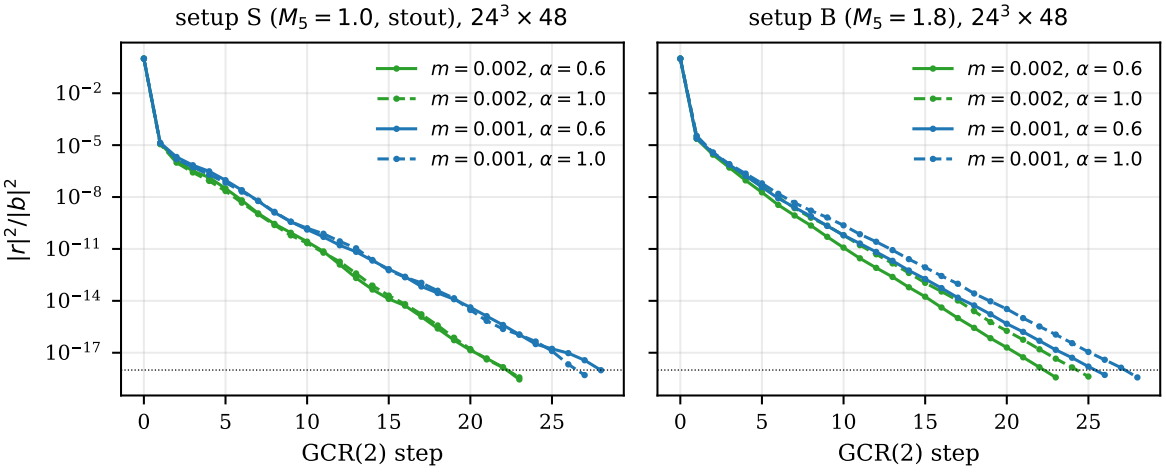


FIG. 6. The GCR(2) residual histories of Table 8, $24^3 \times 48$. `data/gcr_steps_24.tsv`.